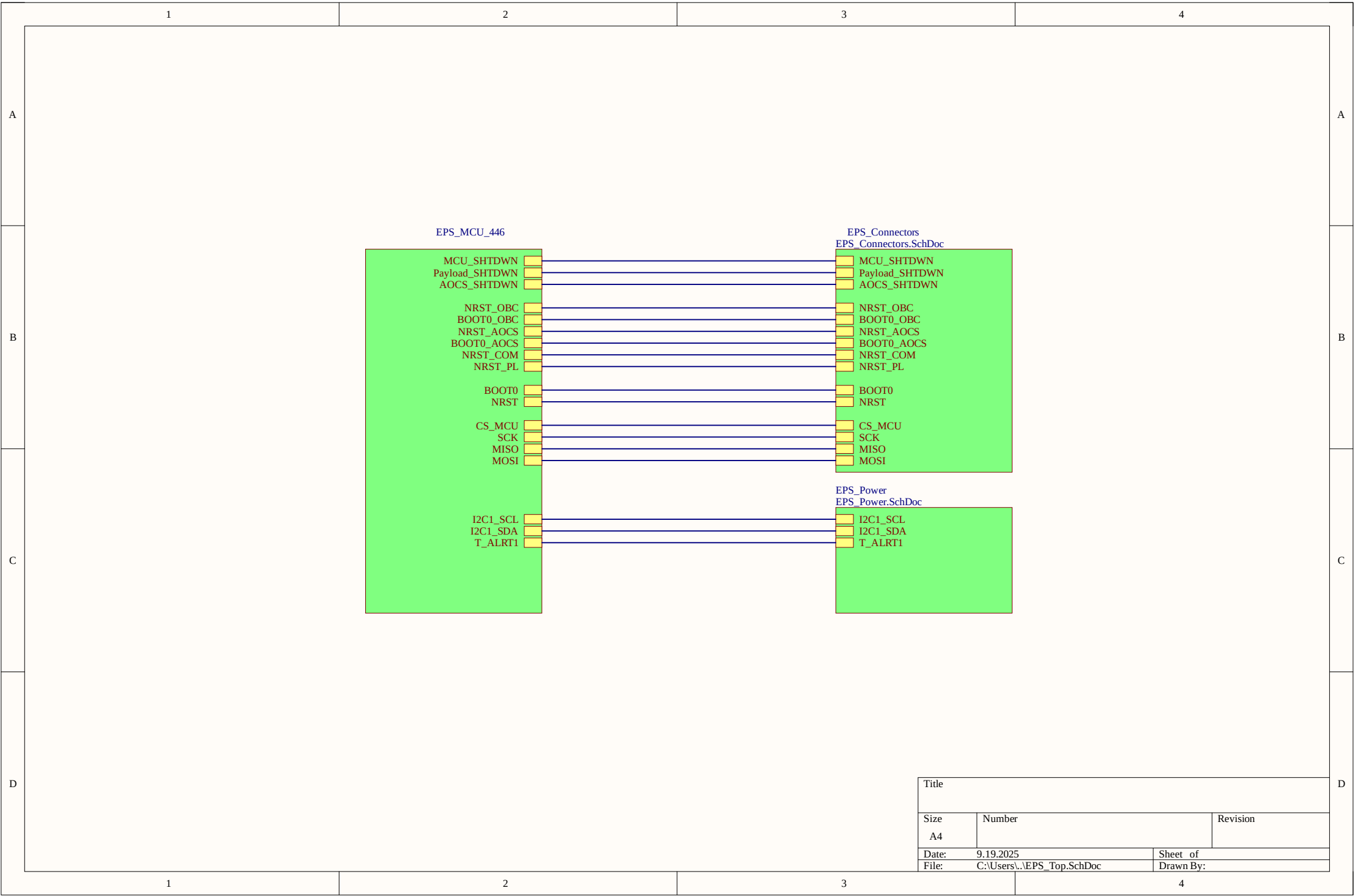
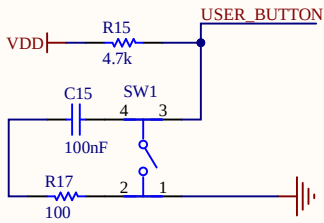


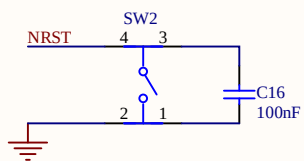


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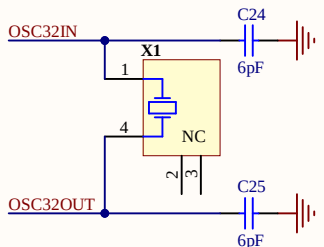
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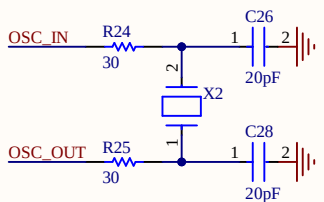
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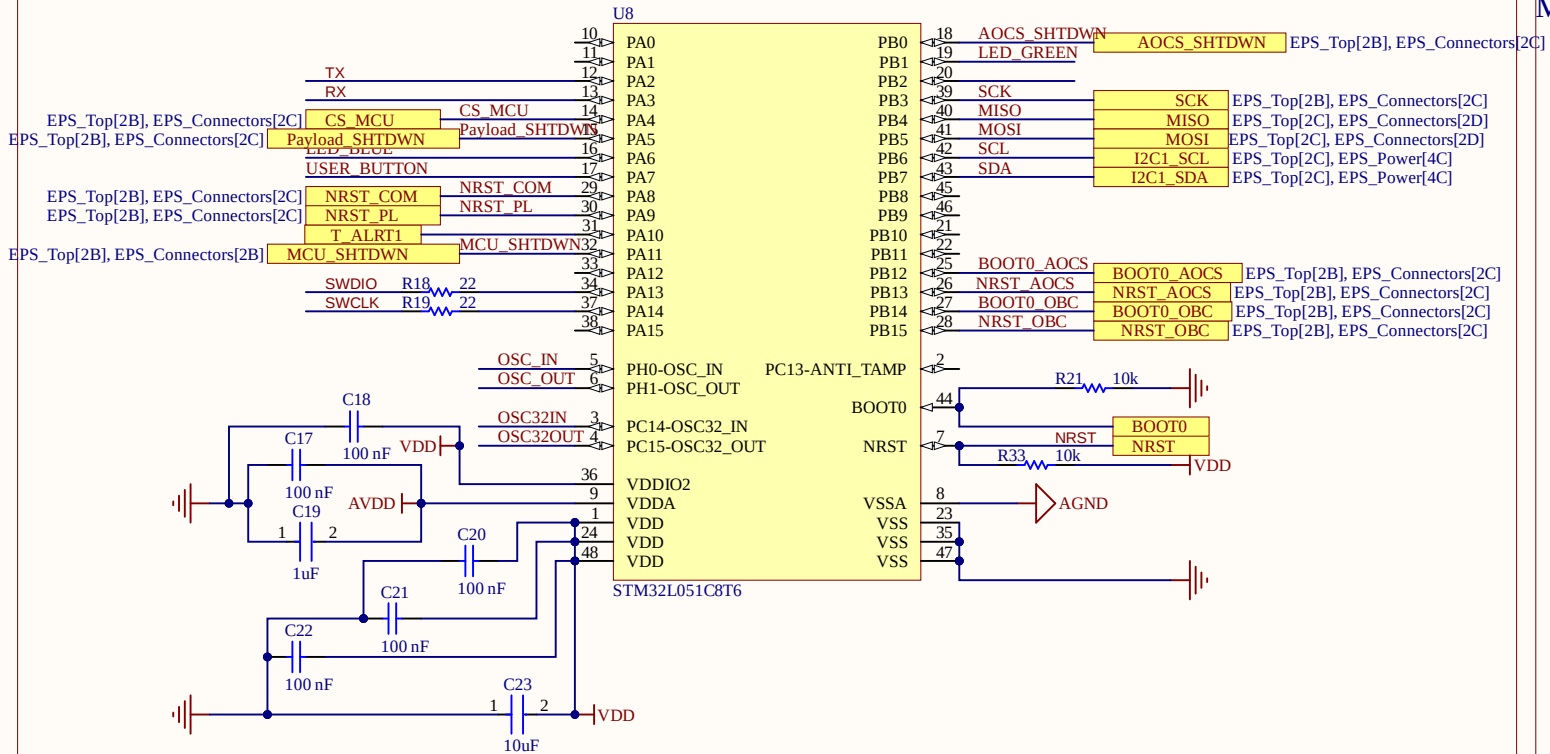
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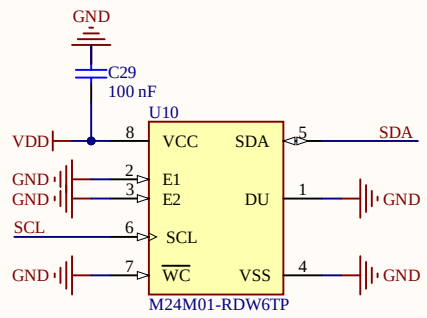
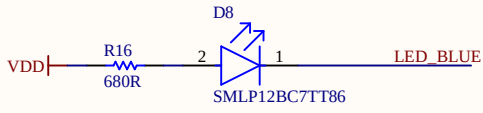
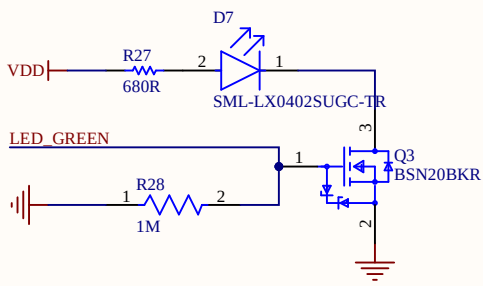
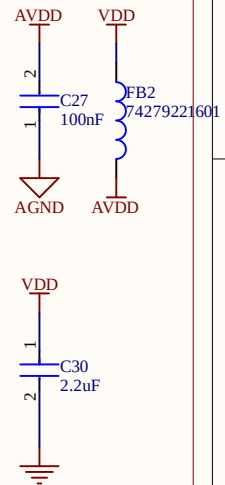
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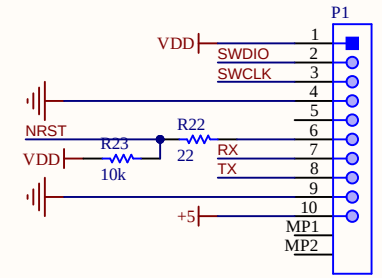
MCU IO



MCU DECAPS

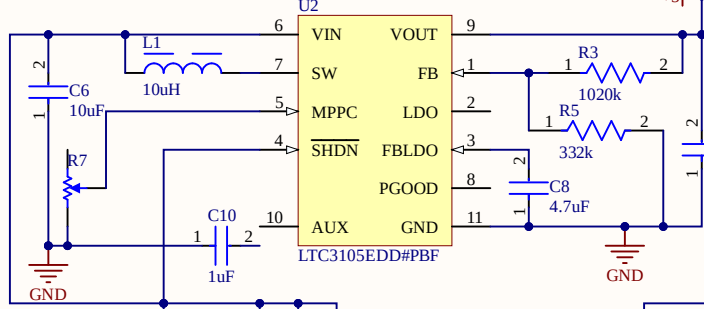


SWD

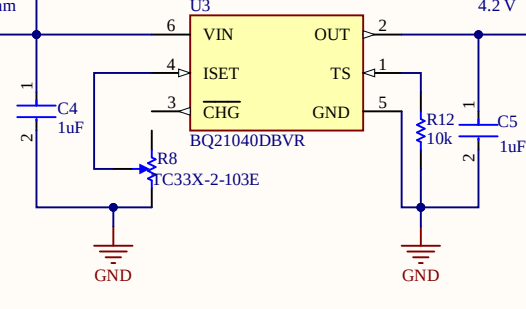


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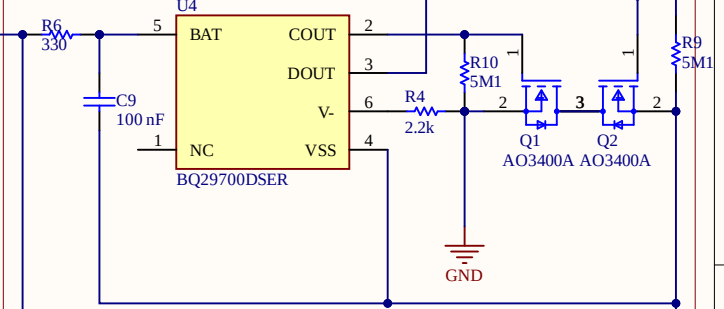
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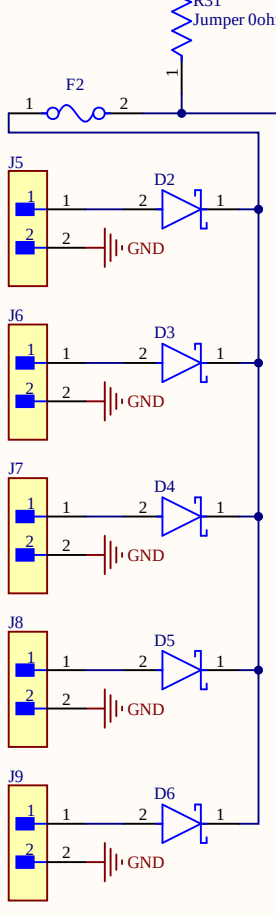
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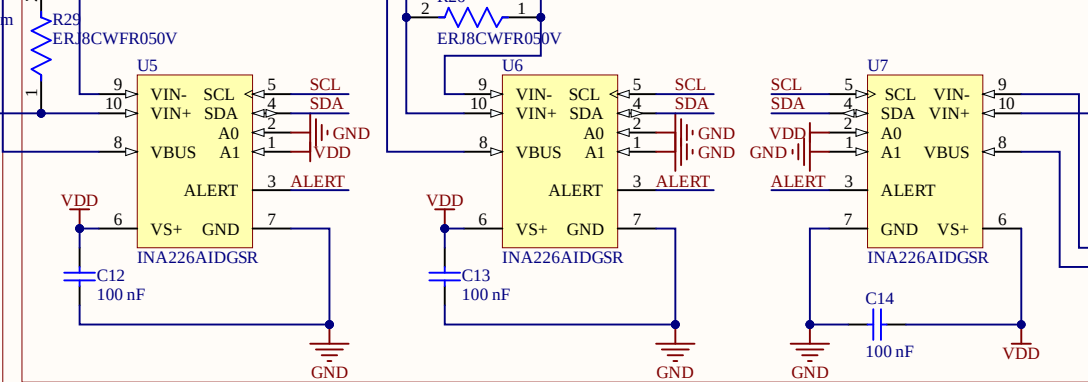
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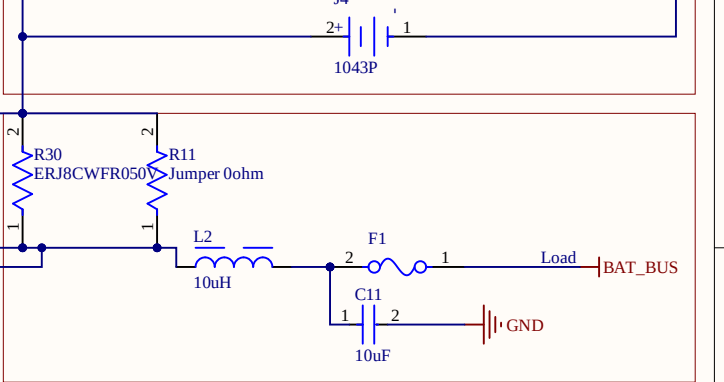
Solar Cells



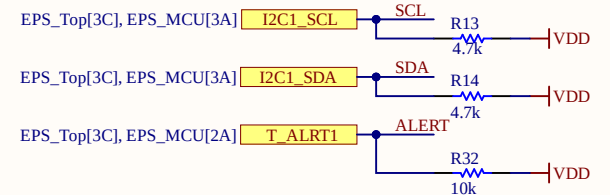
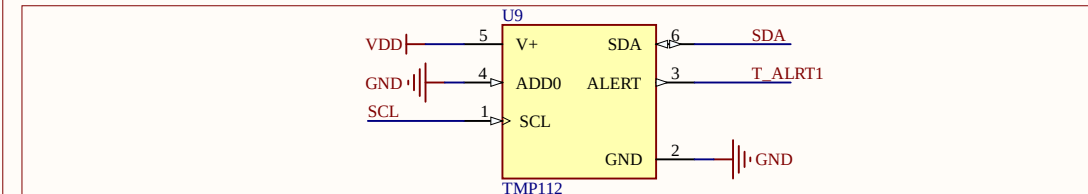
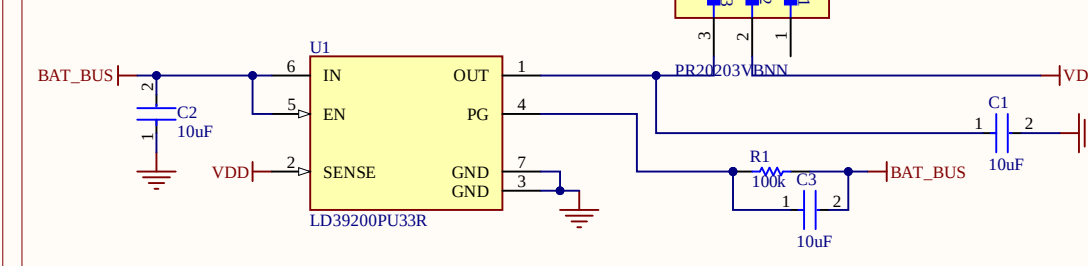
Sensors



Battery



LDO 3V3



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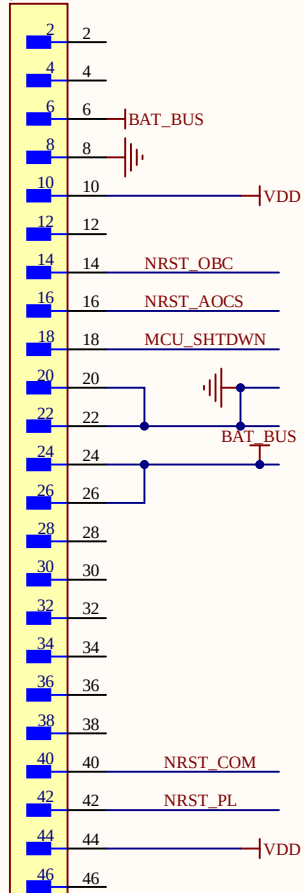
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C

D

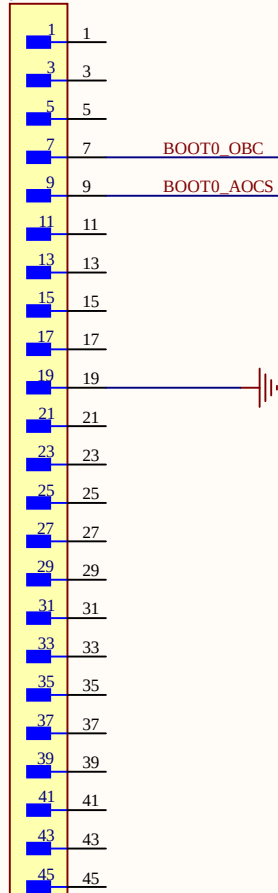
Morpho connector

J2B



PBC23DAFN

J2A



PBC23DAFN

EPS_Top[3B], EPS_MCU[2A] MCU_SHTDWN MCU_SHTDWN
EPS_Top[3B], EPS_MCU[2A] Payload_SHTDWN Payload_SHTDWN
EPS_Top[3B], EPS_MCU[3A] AOCS_SHTDWN AOCS_SHTDWN

EPS_Top[3B], EPS_MCU[3B] NRST_OBC NRST_OBC
EPS_Top[3B], EPS_MCU[3B] BOOT0_OBC BOOT0_OBC
EPS_Top[3B], EPS_MCU[3B] NRST_AOCS NRST_AOCS
EPS_Top[3B], EPS_MCU[3B] BOOT0_AOCS BOOT0_AOCS

EPS_Top[3B], EPS_MCU[2A] NRST_COM NRST_COM
EPS_Top[3B], EPS_MCU[2A] NRST_PL NRST_PL

EPS_Top[3B], EPS_MCU[2A] CS_MCU CS_MCU

EPS_Top[3B], EPS_MCU[3A] SCK SCK

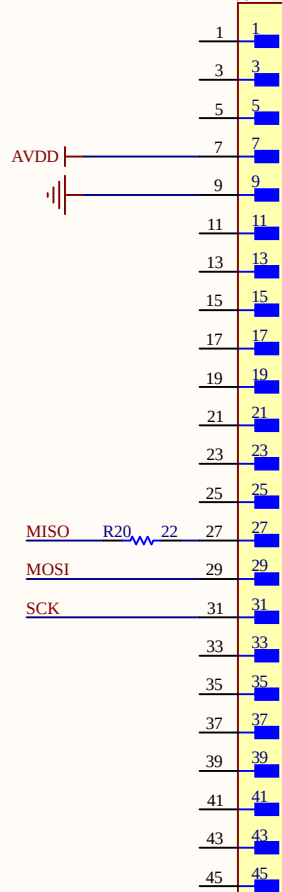
EPS_Top[3C], EPS_MCU[3A] MISO MISO

EPS_Top[3C], EPS_MCU[3A] MOSI MOSI

EPS_Top[3B], EPS_MCU[3B] BOOT0 BOOT0

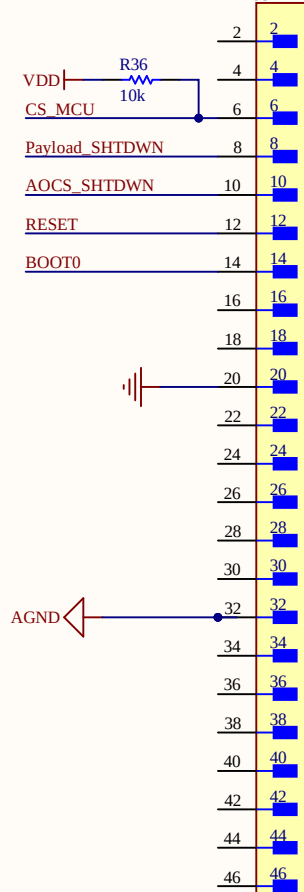
EPS_Top[3B], EPS_MCU[3B] NRST RESET

J3A



PBC23DAFN

J3B



PBC23DAFN

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A

B

C

D

Line #	Name	Description	Designator	Revision ID	Revision State	Revision Status	Quantity	Manufacturer 1	Manufacturer Part Number 1	Manufacturer Lifecycle 1	Supplier 1	Supplier Part Number 1	Supplier Unit Price 1	Supplier Subtotal 1
	10uF	CAP CER 10UF 16V X7R0805	C1, C2, C3, C6, C7, C11, C23		Unknown server		7	Samsung Electro-Mechanics	CL21B106KQNNING	Volume Production	Mouser	187-CL21B106KQNNING	0,087	0,87
	1uF	Cap Ceramic 1uF 25V XSR±10% Pad SMD 0603 +85°C T/R	C4, C5		Unknown server		2	Samsung Electro-Mechanics	CL10A105KA8NNNC	Volume Production	Newark	82AC9275	0,005	0,01
	4.7uF	CAP CER 4.7UF 16V X7R0805	C8		Unknown server		1	Yageo Group	CC0805KXK7R7BB475	Volume Production	Mouser	603-CC0805KXK7R7BB475	0,38	0,38
	100 nF, 100nF	Chip Capacitor, 100 nF, +/- 10%, 16 V, -55 to 125 degC, 0402 (1005 Metric), RoHS, Tape and Reel	C9, C12, C13, C14, C15, C16, C17, C18, C20, C21, C22, C29		Unknown server		12	Samsung Electro-Mechanics	CL05B104K05NNNC	Volume Production	Farnell	3013349	0,00777	0,0933
	1uF	Cap Ceramic 1uF 16V XSR±10% Pad SMD 0603 +85°C T/R	C10, C19		Unknown server		2	Samsung Electro-Mechanics	CL10A105K08NNNC	Volume Production	Mouser	187-CL10A105K08NNNC	0,005	0,05
	6pF		C24, C25		Unknown server		2	Murata	GRM1555C1H6R0DA01D	Not Recommended for New Design	Mouser	81-GRM1555C1H6R0DA1D	0,1	0,2
	20pF	Ceramic Capacitor for Automotive 20pF ±2% 50VDC C0G 0402 Paper T/R	C26, C28		Unknown server		2	Murata	GCM1555C1H200GA16D	Volume Production	Mouser	81-GCM1555C1H200GA6D	0,1	0,2
	100nF	CAP CER 0.1UF 16V X7R0402	C27		Unknown server		1	Yageo Group	CC0402KFK7R7BB104	Volume Production	Mouser	603-CC0402KFK7R7BB104	0,1	0,1
	2.2uF	Cap Ceramic 2.2uF 16V XSR±10% Pad SMD 0603 +85°C T/R	C30		Unknown server		1	Samsung Electro-Mechanics	CL10A225K08NNNC	Volume Production	Mouser	187-CL10A225K08NNNC	0,1	0,1
	BAT60AEB327HTSA1	DIODE SCHOTTKY 10V 3A SOD323-2	D2, D3, D4, D5, D6		Unknown server		5	Infineon	BAT60AEB327HTSA1	Volume Production	Newark	85X4127	0,084	0,42
	SML-LX0402SUGC-TR	LED GREEN CLEAR SMD	D7		Unknown server		1	Lumex	SML-LX0402SUGC-TR	Volume Production	RS(Formerly Allied Electronics)	70127287	0,14	0,14
	SMLP12BC7/T186	LED BLUE CLEAR 1006 SMD	D8		Unknown server		1	ROHM	SMLP12BC7/T186	Obsolete				
	MF-NSMF020-2	PTC RESET FUSE 24V 200MA 1206	F1, F2		Unknown server		2	Bourns	MF-NSMF020-2	Volume Production	RS(Formerly Allied Electronics)	70153457	0,2	0,4
	74279221601	74279221601 Ferrite Beads Multi-Layer High Current 600Ohm 25% 100MHz 2.5A 0.05Ohm DCR 1206 Automotive T/R	FB2		Unknown server		1	Würth Elektronik	74279221601		Newark	40Y3218	0,058	0,058
	PR20203VBNN	CONN HEADER VERT 3POS 2.54MM	J1		Unknown server		1	Metz Connect	PR20203VBNN	Volume Production	Newark	43AC1814	0,108	0,108
	PBC23DAFN	CONN HEADER VERT 48POS 2.54MM	J2, J3		Unknown server		2							
	1043P	BATT HOLDER 18650 1 CELL PC PIN	J4		Unknown server		1	Keystone	1043P	Volume Production	Mouser	534-1043P	2,81	2,81
	B2B-XH-A(LF)(SN)	CONN HEADER VERT 2POS 2.5MM	J5, J6, J7, J8, J9		Unknown server		5	IST	B2B-XH-A(LF)(SN)	Volume Production	Future Electronics	1638840	0,0501	0,2505
	10uH		L1, L2		Unknown server		2	Würth Elektronik	74437324100		Mouser	710-74437324100	2,35	4,7
	SML0B-SHLS-TR(LF)(SN)	Male Header, Pitch 1mm, 1 x 10 Position, Height 1.8mm, -25 to 85 degC	P1		Unknown server		1	IST	SML0B-SHLS-TR(LF)(SN)	Volume Production	Mouser	306-SML0BSHLS-TR(LF)(SN)	0,82	0,82
	AC93400A	MOSFET N-CH 30V 5.7A SOT23	Q1, Q2		Unknown server		2	Alpha & Omega Semiconductor	AC93400A	Volume Production	Arrow Electronics	AC93400A	0,3297	0,6594
	BSN20BKR	MOSFET N-CH 60V 265MA TO236AB	Q3		Unknown server		1	Nexperia	BSN20BKR	Volume Production	Mouser	771-BSN20BKR	0,25	0,25
	100k		R1		Unknown server		1	Yageo Group	RC0402FR-07100KL	Volume Production	Amet	67F8887	0,005	0,005
	Jumper 0ohm	RESSMD 0 OHM JUMPER 1/4W 1206	R2, R11, R31		Unknown server		3	Yageo Group	RC1206JR-070RL	Volume Production	Newark	26M6809	0,017	0,17
	1020k	RES Thick Film, 1.02MQ, 1%, 0.1W, 100ppm/°C, 0603	R3		Unknown server		1	Vishay	CRCW06031M02FKEA	Volume Production	Mouser	71-CRCW06031M02FKEA	0,1	0,1
	2.2k	Thick Film Resistor, 2.2 Kohm, 100 mW, 50 V, +/- 1%, -55 to 155 degC, 2-Pin SMD (0603), RoHS, Tape and Reel	R4		Unknown server		1	Yageo Group	RC0603FR-072K2L	Volume Production	Mouser	803-RC0603FR-072K2L	0,1	0,1
	332k	RES Thick Film, 332kΩ, 1%, 0.1W, 100ppm/°C, 0603	R5		Unknown server		1	Vishay	CRCW0603332FKEA	Volume Production	Newark	25AC7470	0,003	0,003
	330	Chip Resistor, 330 Ohm, +/- 1%, 0.1 W, -55 to 155 degC, 0603 (1608 Metric), RoHS, Tape and Reel	R6		Unknown server		1	Panasonic	ERJ-3BFK3300V	Volume Production	Mouser	667-ERJ-3BFK3300V	0,12	0,12
	TC33X2-103E	Trimming Potentiometer, 10000 Ohm, -40 to 100 degC, 3-Pin SMD, RoHS, Tape and Reel	R7, R8		Unknown server		2	Bourns	TC33X2-103E	Volume Production	Newark	97M6836	0,137	0,274
	5Ml		R9, R10		Unknown server		2	Vishay	CRCW06035M10FKEA	Volume Production	Arrow Electronics	CRCW06035M10FKEA	0,0127	0,0254
	10k		R12, R21, R23, R33		Unknown server		4	Yageo Group	RC0402FR-0710KL	Volume Production				
	4.7k	Chip Resistor, 4.7 Kohm, +/- 1%, 0.1 W, -55 to 155 degC, 0402 (1005 Metric), RoHS, Tape and Reel	R13, R14, R15		Unknown server		3	Panasonic	ERJ-2RFK4701X	Volume Production	Mouser	667-ERJ-2RFK4701X	0,017	0,17
	680R		R16, R27		Unknown server		2	Vishay	CRCW0402680FKEA	Volume Production	Newark	25AC7225	0,011	0,022
	100		R17		Unknown server		1	Panasonic	ERJ-3BFK1000V	Volume Production	Amet	85Y8496	0,099	0,099
	22	Chip Resistor, 22 Ohm, +/- 1%, 0.1 W, -55 to 155 degC, 0603 (1608 Metric), RoHS, Tape and Reel	R18, R19, R20, R22		Unknown server		4	Yageo Group	RC0603FR-0722RL	Volume Production	Newark	26M6754	0,009	0,09
	30		R24, R25		Unknown server		2	Vishay	CRCW0603300FKEA	Volume Production	Newark	25AC7452	0,015	0,03
	ERBCWFR050V	RES 0.05 OHM 1% 1W 1206	R26, R29, R30		Unknown server		3	Panasonic	ERBCWFR050V	Volume Production	Future Electronics	1103980	0,385	1,16
	1M	Thick Film Chip Resistor, 0402, 1MQ, 1%, 100ppm/°C, 0.063W, 50V	R28		Unknown server		1	ROHM	MCR01MR1F1004	Obsolete				
	10k	Chip Resistor, 10 Kohm, +/- 1%, 0.1 W, -55 to 155 degC, 0402 (1005 Metric), RoHS, Tape and Reel	R32, R36		Unknown server		2	Panasonic	ERJ-2RFK1002X	Volume Production	Mouser	667-ERJ-2RFK1002X	0,1	0,2
	PTS810SM250SMTRLFS	SWITCH TACTILE SPST-NO 0.05A 16V	SW1, SW2		Unknown server		2	C&K Components	PTS810SM250SMTRLFS	Volume Production	Newark	35AH5295	0,074	0,148
	LD39200PU33R	IC REG LINEAR 3.3V 2A 6DFN	U1		Unknown server		1	STMicroelectronics	LD39200PU33R	Volume Production	Newark	29AC7323	0,4	0,4
	LTC3105EDD#PBF	400 mA Step-Up DC/DC Converter with Maximum Power Point Control and 250 mV Start-Up, 10-pin SON (DD-10), -40 to 85 degC, Pb-Free	U2		Unknown server		1	Analog Devices	LTC3105EDD#PBF	Volume Production	Arrow Electronics	LTC3105EDD#PBF	1,99	1,99
	BQ21040DBVR	IC BATT CHG LI-ION 1CELL SOT23-6	U3		Unknown server		1	Texas Instruments	BQ21040DBVR	Volume Production	Mouser	595-BQ21040DBVR	0,88	0,88
	BQ29700DSER	IC LITHIUM BATT PROTECT 6-SON	U4		Unknown server		1	Texas Instruments	BQ29700DSER	Volume Production	Arrow Electronics	BQ29700DSER	0,09	0,09
	INA226AIDGSR	High-Side Measurement, Bi-Directional Current / Power Monitor with I2C Interface, 2.7 to 5.5 V, -40 to 125 degC, 10-pin SOP (DGS10), Green (RoHS & no Sb/Br)	U5, U6, U7		Unknown server		3	Texas Instruments	INA226AIDGSR	Volume Production				
	STM32L051C8T6	Ultra-Low-Power ARM Cortex-M0+ MCU, 64 KB Flash, 1.8 to 3.6 V, -40 to 85 degC, 48-Pin LQFP, RoHS, Tray	U8		Unknown server		1	STMicroelectronics	STM32L051C8T6	Volume Production	Newark	83X7779	2,2	2,2
	TMP112	SENSOR DIGITAL -40C-125C 6SOT TMP112AIDRJR, NOT AQDRQR1	U9		Unknown server		1	Texas Instruments	TMP112AQDRQR1	Volume Production	Mouser	595-TMP112AQDRQR1	1,91	1,91
	M24M01-RDM6TP	IC EEPROM 1M I2C 1MHZ 8TSSOP	U10		Unknown server		1	STMicroelectronics	M24M01-RDM6TP	Not Recommended for New Design	Newark	26AM9329	0,554	0,554
	ABS25-32.768KHZ-6-T	Low Frequency Crystal, 32.768 KHz, 6 pF, +/- 20 ppm, -40 to 85 degC, 4-Pin SMD, RoHS, Tape and Reel	X1		Unknown server		1	Abracon	ABS25-32.768KHZ-6-T	Volume Production	Newark	60F9393	0,467	0,467
	ABM3-8.000MHZ-D2Y-T	Crystal 8MHz ±20ppm 18pF SMD 2.5mm x 3.2mm	X2		Unknown server		1	Abracon	ABM3-8.000MHZ-D2Y-T	Volume Production	RS(Formerly Allied Electronics)	71528075	0,256	0,256